

Rotary Moulder Workbench

User Guide

A FreeCAD workbench for designing rotary cookie moulder drums. Build the cylindrical drums used in industrial biscuit-and-cracker production to form raw dough into shaped cookies.

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1. Concepts

The mould drum

A rotary moulder drum is a cylinder with cookie-shaped cavities cut into its outer surface. Dough is pressed against the drum, fills the cavities, then the formed cookies are released onto a conveyor belt.

For cookies to release properly, every cavity wall needs **draft** — the floor of the cavity is narrower than the rim. This workbench handles the draft automatically.

Flat sketches → curved geometry via Sketch_On_Surface

You design the cookie outline as a flat 2D sketch on the **XY plane**. Before passing it to this workbench, you project the sketch onto the drum's cylindrical face using the **Curves Workbench's Sketch_On_Surface** feature. The resulting Sketch_On_Surface object is what you select as the cookie outline for the cavity.

The mapping convention:

- Sketch **X** coordinate maps to angular position around the drum
- Sketch **Y** coordinate maps to axial position along the drum
- Full sketch width = one full wrap (2π) around the drum, so a sketch X range of $0..(\pi \times \text{diameter})$ corresponds to a full circumference

Detail and docker outlines, by contrast, use flat sketches directly (no Sketch_On_Surface needed) — they are projected internally using the parent cavity's SoS mapping. Use familiar Sketcher tools (rectangles, splines, ShapeString text, points, etc.) for these.

Object hierarchy

A typical project tree looks like:

```
Drum
+-- Cavity (or CavityPattern)
    +-- Outline (a Sketch_On_Surface object)
        |   +-- (source flat sketch on XY plane)
    +-- CavityDetail (letters/shapes inside the cavity)
        |   +-- Outline (flat sketch on XY plane)
    +-- CavityDockers (perforation pins)
        +-- Outline (flat sketch on XY plane with points)
```

Each cavity can have multiple detail and docker children. The cavity outline is an SoS object; detail and docker outlines are flat sketches.

Draft direction

All cavities use **floor_narrower** draft — the cavity opens wider at the drum surface than at the floor, so dough pulls out cleanly. For details (letters/shapes), the same logic applies:

- **Engrave** mode (text recessed into floor): a small drafted pocket inside the cavity, narrower at the bottom of the engraving.
- **Emboss** mode (text raised above floor): a small drafted bump, narrower at the top of the bump.

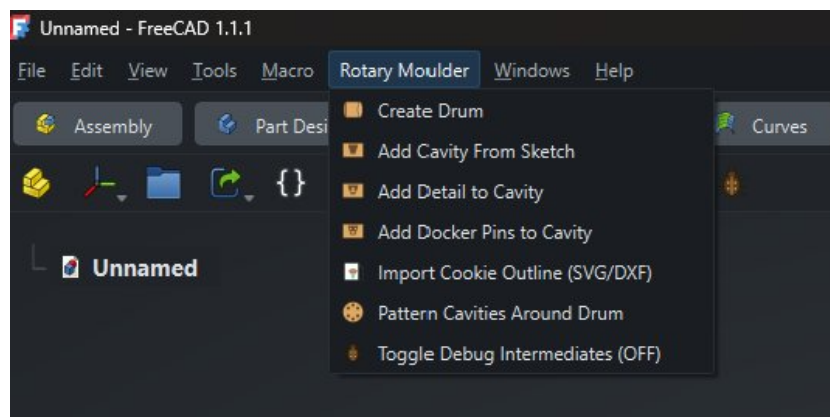
In both modes the wider side is whichever is outside the cookie, so the dough releases.

2. Tutorial: build a complete mould

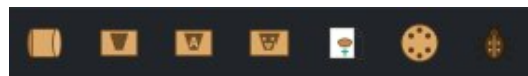
This walkthrough builds a 6×3 pattern of rectangular cookies with the letters MSC embossed and a few docker pins per cookie.

Step 1 — Create the drum

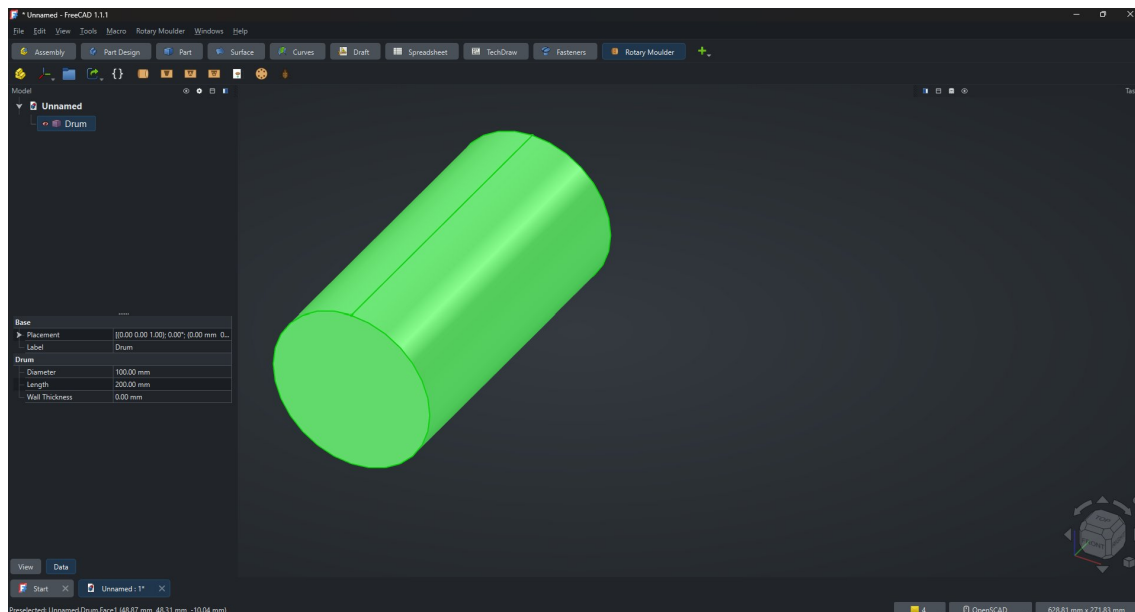
- Switch to the **Rotary Moulder** workbench
- Click **Create Drum**
- Accept defaults (D = 100 mm, L = 200 mm) — or set your own
- A Drum object appears in the tree



Rotary Moulder menu with all available commands



Toolbar buttons



Drum created in tree and 3D viewport

You can change *Diameter* and *Length* anytime; everything else updates automatically.

Step 2 — Sketch the cookie outline (flat)

The sketch goes on the **XY plane**. Think of X as wrapping around the drum; Y runs along the drum's length.

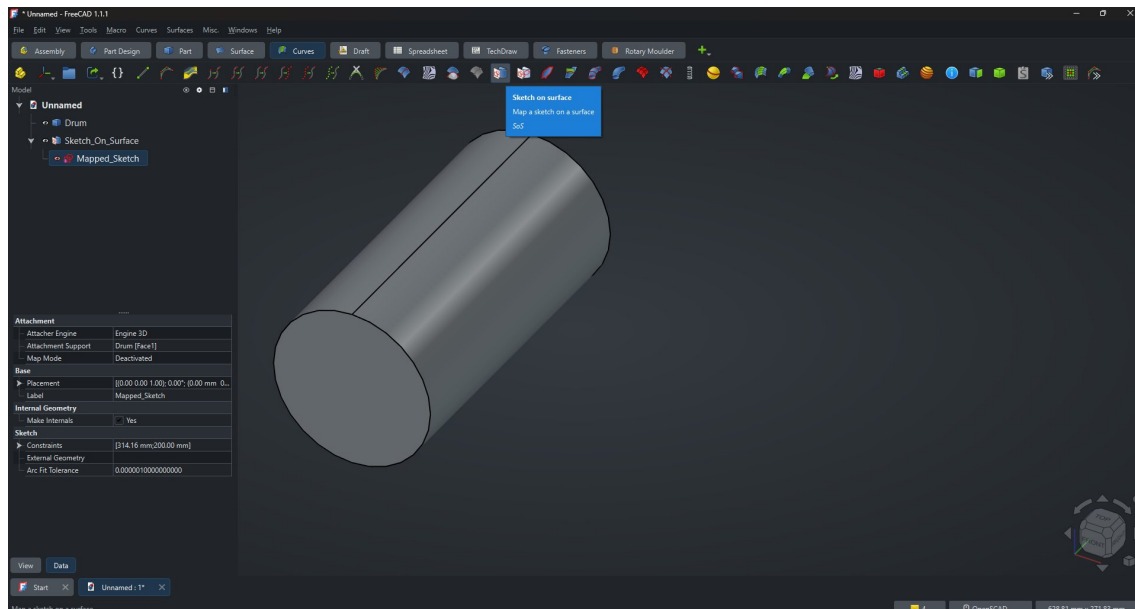
- Right-click tree → Create sketch → XY_Plane
- Draw a rectangle (e.g. 12 × 25 mm) somewhere not at the origin
- Close the sketch (Escape)

Tip: keep your sketch coordinates inside the cookie outline region of the imaginary unwrapped drum. Y must be within the drum length; X within ($\pi \times \text{diameter}$), the circumference.

Step 3 — Project the sketch onto the drum (Sketch_On_Surface)

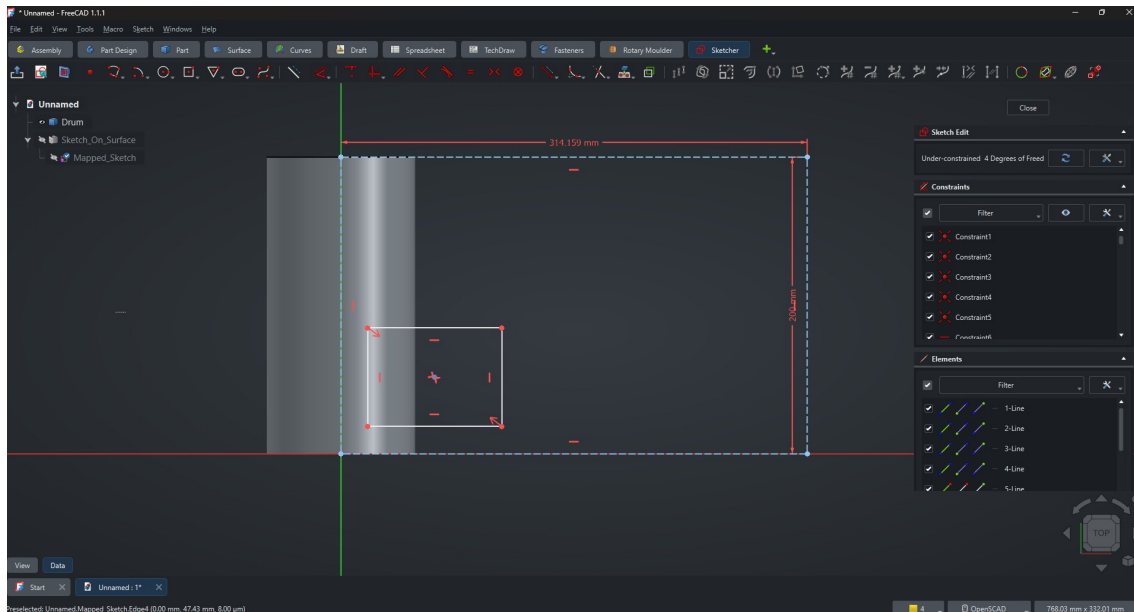
The cavity command expects an outline that has already been wrapped around the drum's cylinder. Use the **Curves Workbench**:

- Switch to the **Curves Workbench**
- Select the drum's outer cylindrical face, then Ctrl-click your flat sketch
- Click **Curves** → **Sketch on Surface**
- A Sketch_On_Surface object (often labeled SoS) appears in the tree, showing your sketch projected onto the drum



Sketch on Surface tool in the Curves Workbench

- Double-click the **Mapped_Sketch** under the SoS object — this opens the Sketcher with a 314.16 × 200 mm boundary box representing the unwrapped drum surface. Draw your cookie outline (e.g. a 12 × 25 mm rectangle) inside this boundary.

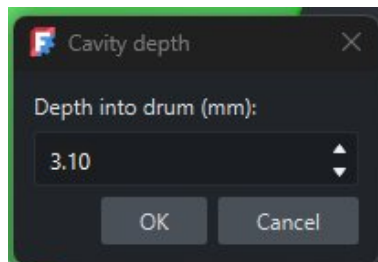


Mapped_Sketch with a rectangle inside the unwrapped boundary

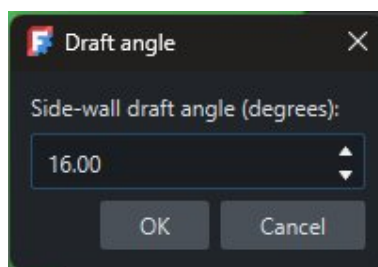
Switch back to the Rotary Moulder workbench.

Step 4 — Add the cavity

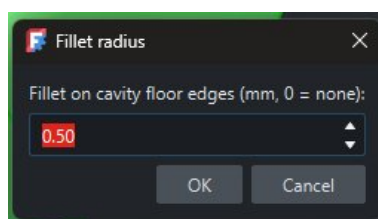
- Select the **drum** AND the **Sketch_On_Surface** (Ctrl+click in tree)
- Click **Add Cavity From Sketch**
- Three dialogs appear in sequence:



Cavity depth dialog

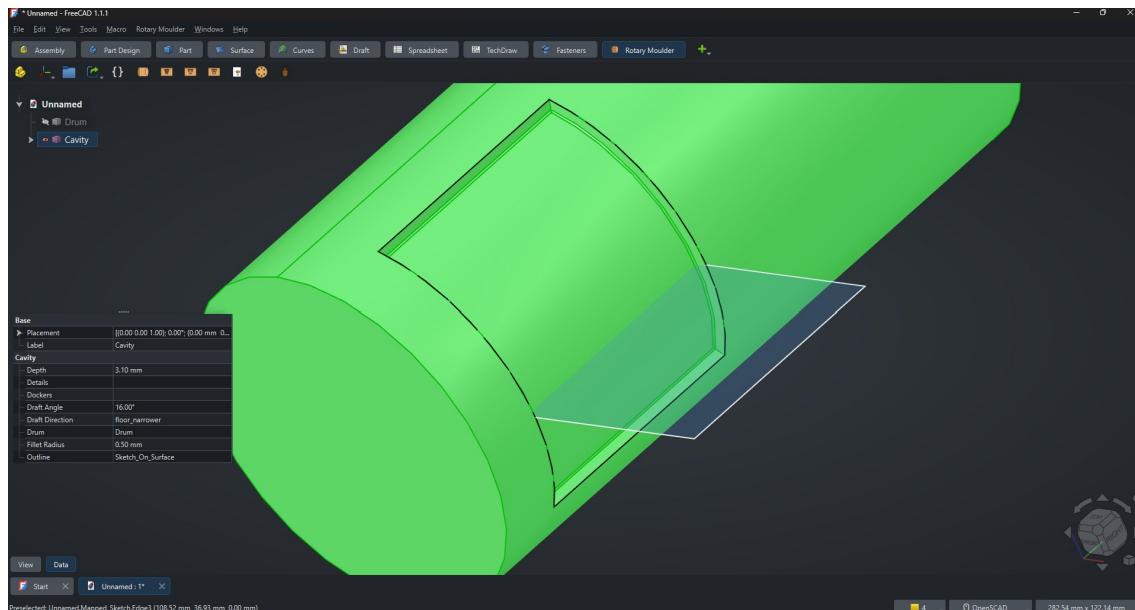


Draft angle dialog



Fillet radius dialog

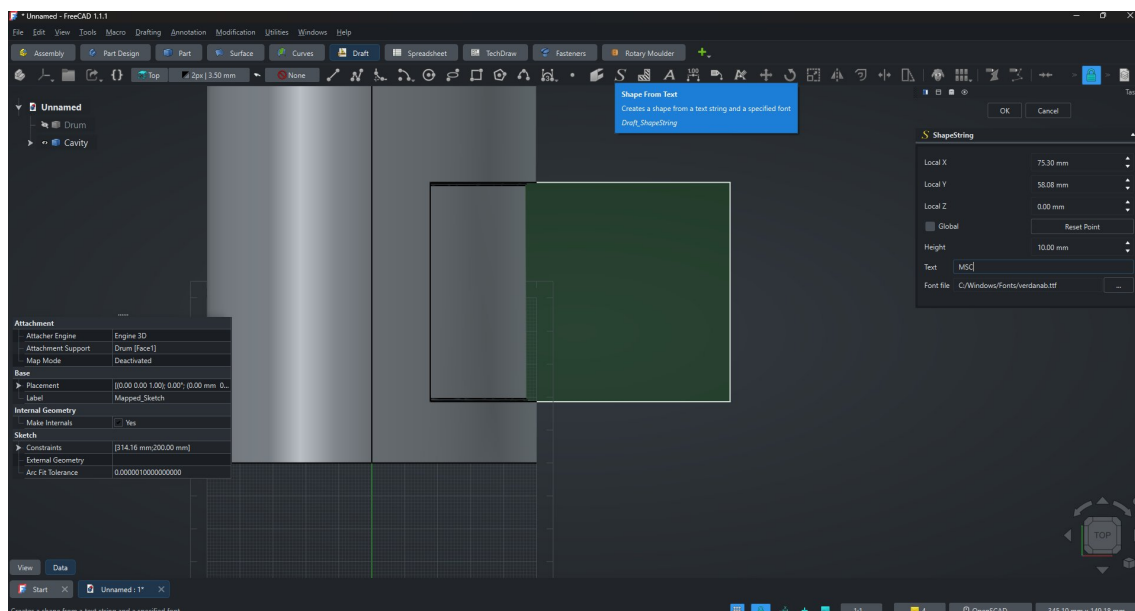
A Cavity object appears under the drum; the drum now shows the cavity carved into it. The SoS becomes a child of the cavity.



Cavity carved into drum, with parameters visible in the property panel

Step 5 — Add a detail (text)

- Make another **flat sketch** on the XY plane (no SoS this time — details are projected internally)
- Use **Sketcher** → **ShapeString** to add the text MSC. **Tested font: verdanab.ttf** (Verdana Bold) — gives reliable letter shapes
- Position the ShapeString inside the cookie outline area

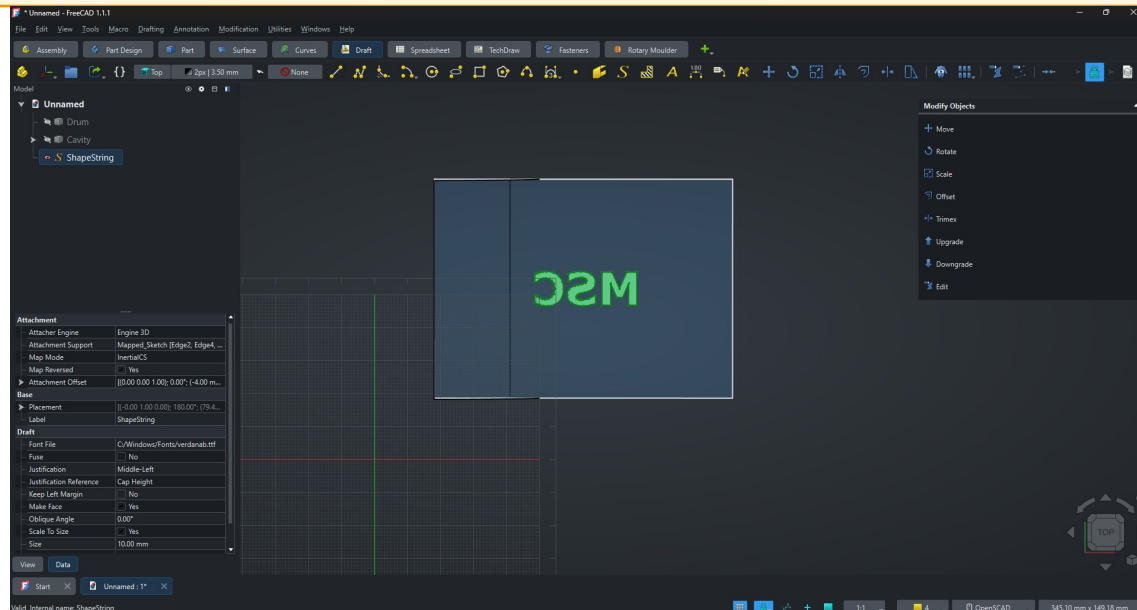


ShapeString creation dialog with verdanab.ttf font

■ Important: mirror the text via InertialCS + Map Reversed

Because the cookie is formed by the drum's **NEGATIVE** volume (the cavity), text needs to be **mirrored** on the drum so it reads correctly on the formed cookie. Otherwise your text comes out reversed in the final product.

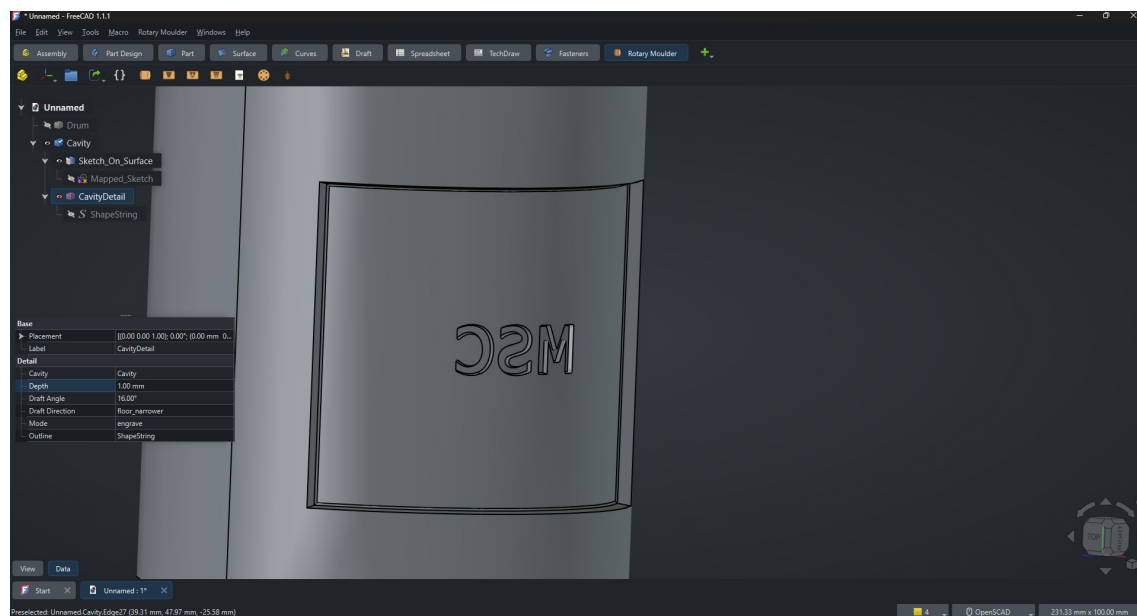
Fix: attach the ShapeString to the SoS's **Mapped_Sketch** with **Map Mode = InertialCS** and **Map Reversed = Yes**. The text will look BACKWARDS in the FreeCAD viewport — that is correct! It will read normally on the produced cookie.



ShapeString attached with InertialCS + Map Reversed (text appears mirrored on drum, which is correct for the cookie orientation)

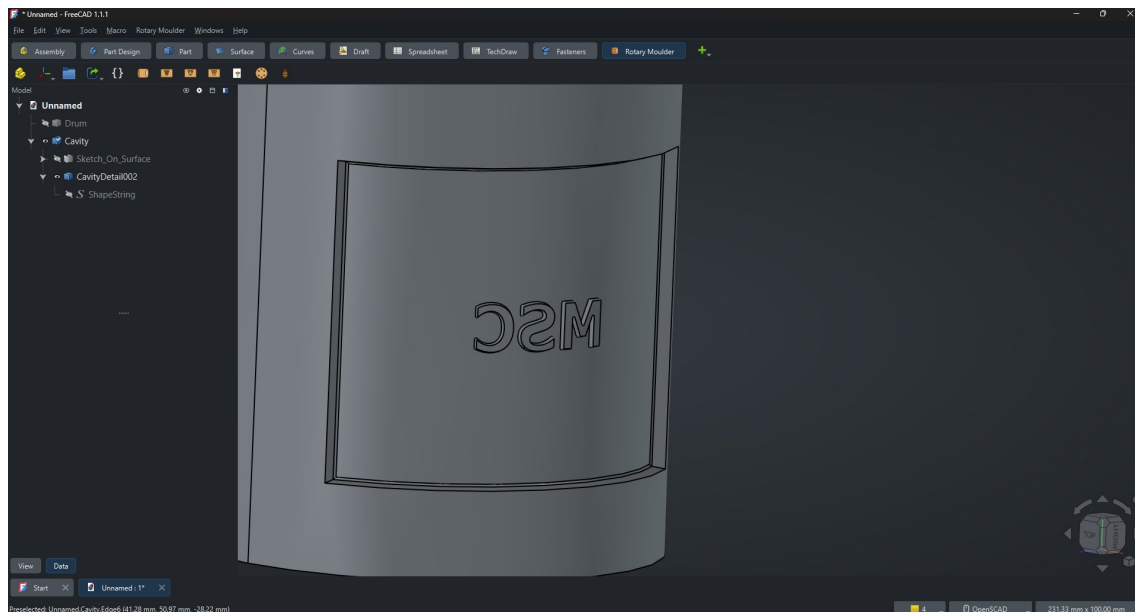
- Close the sketch
- Select the **cavity** AND the **ShapeString**
- Click **Add Detail to Cavity**
- Choose mode (emboss / engrave), depth 0.5 mm, draft angle 16°

The text appears on the cavity floor. With the mirror applied via InertialCS+Map Reversed, it shows mirrored ON the drum — which means correct ON the formed cookie.



Engraved MSC text on cavity floor — mode = engrave

Switching to emboss mode (or using a second detail object with emboss mode) produces raised text instead:

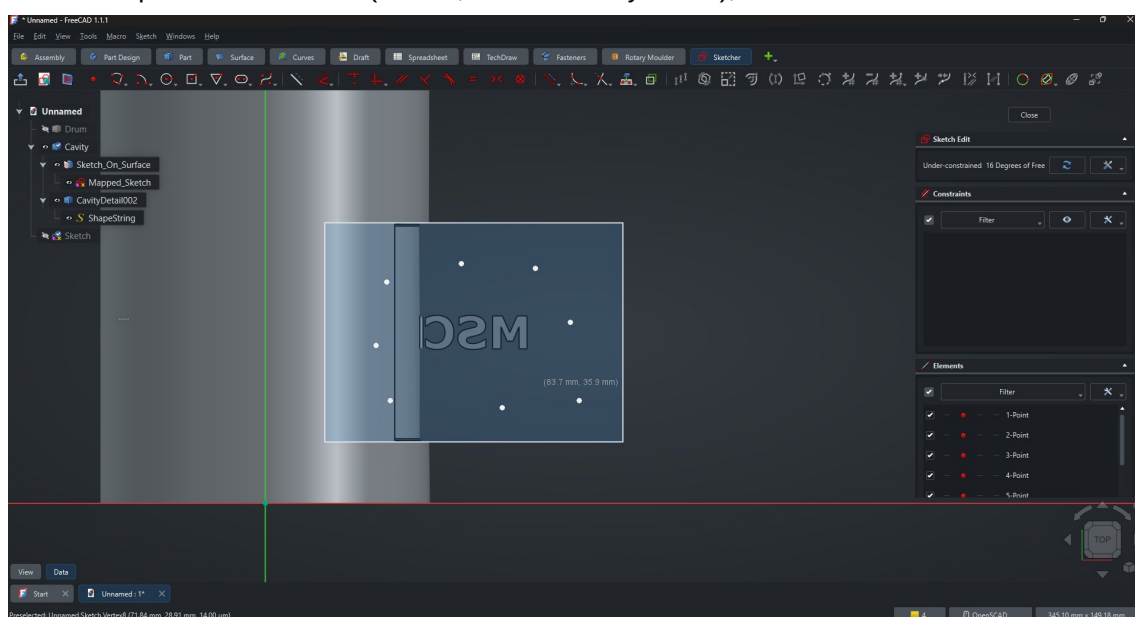


Cavity with embossed MSC after full recompute

Step 6 — Add docker pins

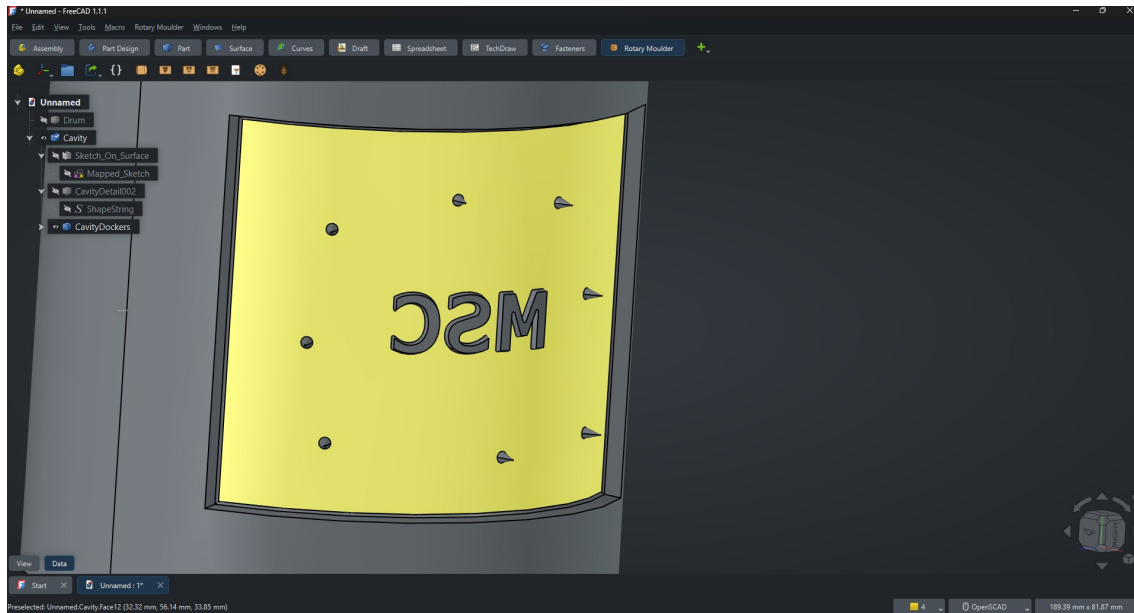
Docker pins are perforations that pierce through the entire cookie thickness, like the dots on a cracker.

- Make another **flat sketch** on the XY plane (also use **InertialCS** + **Map Reversed** to mirror the pin positions to match the cookie orientation)
- Use **Sketcher** → **Create point** to place a few points within your cookie outline area
- Close the sketch
- Select the **cavity** AND the **points sketch**
- Click **Add Docker Pins to Cavity**
- Choose tip diameter 0.2 mm (default, for cracker-style dots), draft 16°



Sketch with points placed for docker pin positions

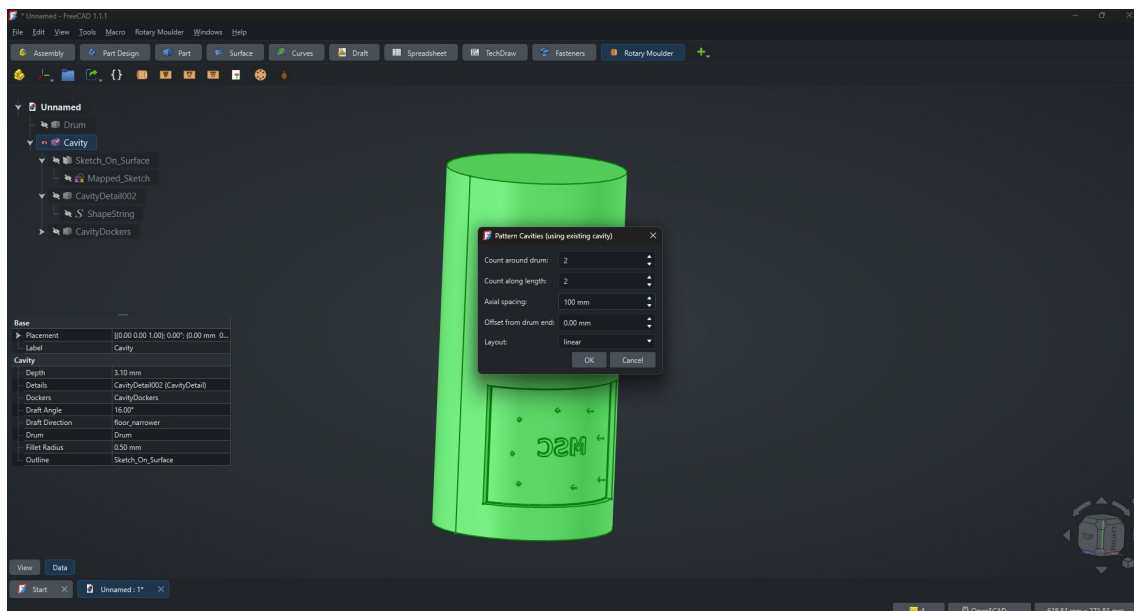
Each point becomes a small mushroom-shaped pin protruding from the cavity floor toward the drum surface.



Cavity with engraved text and docker pins. The yellow floor is the cavity bottom; pins protrude upward and the letters are engraved into the floor.

Step 7 — Pattern around the drum

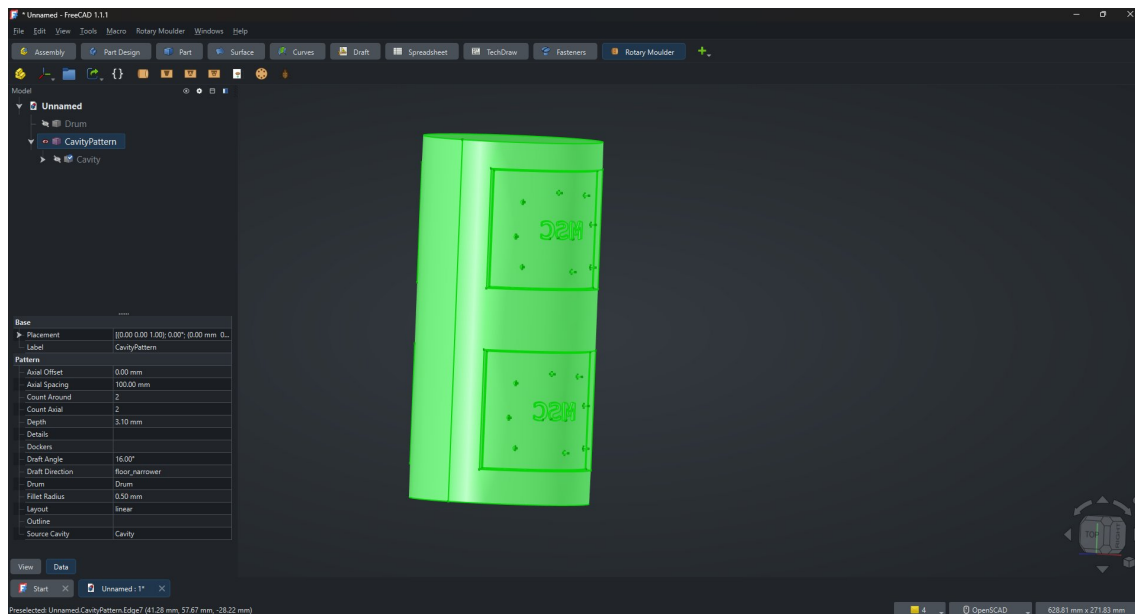
- Select the **drum** AND the **cavity**
- Click **Pattern Cavities Around Drum**
- Set count around (6), count axial (3), spacing, axial offset, layout



Pattern dialog in cavity-inheritance mode (only pattern fields shown)

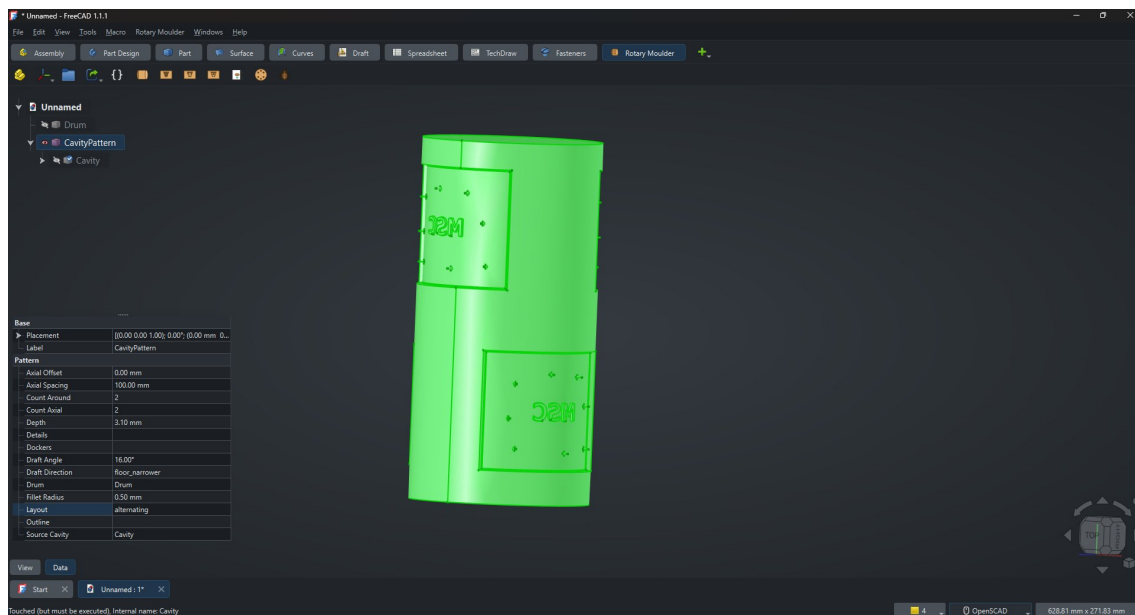
A CavityPattern object appears. The original cavity becomes its source — every cavity instance inherits the source's details and docker pins automatically.

Linear layout:



Linear layout - cavities aligned in a grid

Alternating layout (each row angularly offset by half the step):



Alternating layout - brick-style stagger

Step 8 — Export

Once the drum looks right, use FreeCAD's standard export tools (File → Export As...) to save it as STEP / STL / etc. for manufacturing.

3. Command reference

Create Drum

Creates a cylindrical drum at the origin, axis along Y.

Property	Default	Notes
Diameter	100 mm	Outer diameter of drum
Length	200 mm	Drum length along Y

Add Cavity From Sketch

Cuts a drafted cavity into the drum. Requires the drum + a **Sketch_On_Surface** object (the cookie outline projected onto the drum's cylindrical face via the Curves Workbench) selected.

Field	Default	Notes
Depth	3.1 mm	Cavity depth (cookie thickness)
Draft angle	16 deg	Side-wall draft for release
Fillet radius	0.5 mm	Rounding between floor and walls

The workbench also accepts a plain flat sketch as fallback (via an older code path), but **Sketch_On_Surface** is **strongly recommended** for accurate projection onto the drum.

Add Detail to Cavity

Adds engraved or embossed text/shapes to the cavity floor. Requires a cavity (or pattern) + flat sketch on the XY plane selected. The sketch is projected internally using the parent cavity's SoS mapping.

Field	Default	Notes
Mode	engrave	engrave (recessed) or emboss (raised)
Depth	0.5 mm	Detail depth/height
Draft angle	16 deg	Detail draft for release

A single detail can contain multiple shapes — e.g. one sketch with the text MSC produces three letter chunks treated together.

ShapeString font: the workbench has been tested with **verdanab.ttf** (Verdana Bold). Other fonts may work but some have complex letter shapes that can produce volume-estimation warnings in the report view (geometry is still correct).

Add Docker Pins to Cavity

Adds perforation pins to the cavity floor. Pin tips reach the drum surface, so the formed cookie has pin-prick holes all the way through.

Requires a cavity + sketch containing **points** (Sketcher's Create Point tool) selected.

Field	Default	Notes
Tip diameter	0.2 mm	Diameter at top (dome-end)
Draft angle	16 deg	Side-wall draft of pin

Pin height equals the cavity depth automatically. Base diameter is computed from the draft angle and depth.

Pattern Cavities Around Drum

Replicates a cavity (with its details and dockers) around the drum. Requires either:

- **Drum + Cavity** selected — uses the cavity's existing settings (depth, angle, etc.) and only asks for pattern parameters
- **Drum + Sketch** selected — also asks for cavity parameters and builds the pattern directly from the sketch

Field	Default	Notes
Count around	6	Cavities around the drum perimeter
Count axial	3	Rows along the drum length
Spacing	50 mm	Distance between axial rows
Axial offset	25 mm	Y position of the first row
Layout	linear	linear or alternating (brick-like)

Import Cookie Outline (SVG/DXF)

Imports an SVG or DXF file as a flat sketch that can be used as a cookie outline or detail.

Toggle Debug Intermediates

Toggles whether the workbench saves intermediate construction shapes to an RM_Debug group in the tree. Useful when geometry construction fails. State is persistent across FreeCAD sessions; the button label may not refresh until you switch workbenches.

4. Object reference

Drum

Property	Type	Default	Notes
Diameter	Length	100 mm	Outer drum diameter
Length	Length	200 mm	Drum length along Y

DraftedCavity

Property	Type	Default	Notes
Drum	Link	(required)	Parent drum
Outline	Link	(required)	Sketch_On_Surface projecting a flat sketch
Depth	Length	3.1 mm	Cavity depth
DraftAngle	Angle	16 deg	Side-wall draft
DraftDirection	Enum	floor_narrower	Only option
FilletRadius	Length	0.5 mm	Floor-wall fillet (0 disables)
Details	LinkList	[]	CavityDetail children
Dockers	LinkList	[]	CavityDockers children

CavityDetail

Property	Type	Default	Notes
Cavity	Link	(required)	Parent cavity (or pattern)
Outline	Link	(required)	Flat sketch on XY plane
Depth	Length	0.5 mm	Detail depth or height
DraftAngle	Angle	16 deg	Draft for release
DraftDirection	Enum	floor_narrower	Only option
Mode	Enum	engrave	engrave or emboss

CavityDockers

Property	Type	Default	Notes
Cavity	Link	(required)	Parent cavity (or pattern)
Outline	Link	(required)	Flat sketch on XY plane with point vertices
TipDiameter	Length	0.2 mm	Diameter at pin top
DraftAngle	Angle	16 deg	Pin side-wall draft

CavityPattern

Property	Type	Default	Notes
Drum	Link	(required)	Parent drum
SourceCavity	Link	(optional)	If set, inherits from it
Outline	Link	(optional)	Used when SourceCavity is None

Depth	Length	3.1 mm	When SourceCavity is None
DraftAngle	Angle	16 deg	When SourceCavity is None
FilletRadius	Length	0.5 mm	When SourceCavity is None
CountAround	Integer	6	Cavities around the drum
CountAxial	Integer	3	Rows along the drum
Spacing	Length	50 mm	Axial spacing between rows
AxialOffset	Length	25 mm	First row Y position
Layout	Enum	linear	linear or alternating
Details	LinkList	[]	Extra details on the pattern
Dockers	LinkList	[]	Extra docker groups on the pattern

5. Patterns

Two ways to define a pattern

From a Cavity (recommended): make one cavity, add details and dockers to it, then pattern it. The pattern inherits everything from the source cavity and you only need to set how many copies.

From a Sketch: skip the standalone cavity and define the pattern directly. Useful for simple patterns without details.

How the pattern is built

For performance, the workbench builds the cavity-with-details once at the origin (the master), then copies and rotates it to each pattern position and does a single boolean cut. Much faster than building each cavity independently.

Linear vs alternating layout

- **Linear:** cavities aligned in a perfect grid
- **Alternating:** every other row is angularly offset by half the step (brick-like). Good for packing more cookies on the drum.

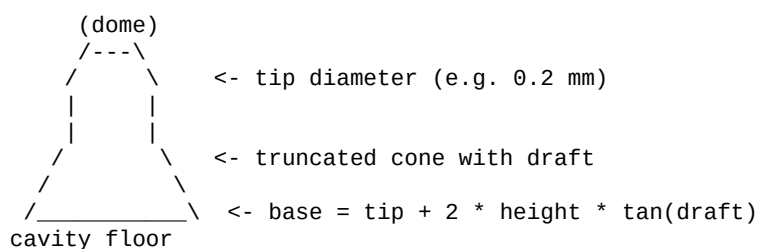
Adding details to the pattern

You can attach additional details or dockers directly to the CavityPattern (in addition to those inherited from the SourceCavity). These apply to every cavity instance.

6. Docker pins

Anatomy of a pin

Each pin is a truncated cone with a hemispherical (dome) tip:



The pin's tip sits just below the drum's outer surface (0.05 mm clearance for boolean reliability). In practice this means the formed cookie has a pin-prick going almost all the way through.

How positions are specified

The outline sketch should contain **points** (Sketcher → Create Point) at each desired pin location. Each vertex in the sketch becomes one pin.

Default parameters

The default 0.2 mm tip is sized for cracker-style perforations. For larger biscuit perforations try 0.5 - 1 mm. The base diameter scales automatically with the depth and draft angle.

7. Debug mode

When geometry construction fails (or just to understand what's happening), enable debug mode by clicking the **Toggle Debug Intermediates** button. Intermediate shapes are saved into a `RM_Debug` group:

Shape name pattern	What it is
<code>RM_cav_rim_wire</code>	Cookie outline projected onto drum surface
<code>RM_cav_floor_wire</code>	Cookie outline at cavity floor (scaled inward)
<code>RM_cav_side_shell</code>	Loft surface between rim and floor wires
<code>RM_cav_top_cap</code>	Top closing face (on drum surface)
<code>RM_cav_bot_cap</code>	Bottom closing face (on cavity floor)
<code>RM_cav_solid</code>	Final cavity-chunk solid
<code>RM_det_fN_outer_*</code>	Detail face N: chunks/wires/caps/solid

Debug state persists across FreeCAD sessions. Toggle off for cleaner operation in production work.

8. Troubleshooting

Text in details comes out backward on the formed cookie

When the cookie is formed by the drum's negative space, anything on the drum is mirrored relative to the cookie. To compensate:

- Attach the ShapeString to the SoS object's **Mapped_Sketch**
- Set **Map Mode = InertialCS**
- Set **Map Reversed = Yes**

This pre-mirrors the text on the drum so the formed cookie reads the right way around. The text will look backward in the FreeCAD viewport — that is correct! It will read normally on the produced cookie. The same principle applies to docker pin positions if they should match a specific orientation.

The cavity is huge / misshapen

Common causes:

- **Missing Sketch_On_Surface:** the cavity outline should be a `Sketch_On_Surface` object, not the flat sketch itself. The SoS object is what tells the workbench where on the drum the outline belongs. If you select the flat sketch directly, the workbench falls back to a less-accurate projection that may misposition the cavity.
- **Sketch placement:** the source flat sketch must be on the **XY plane** with the outline in positive coordinates within the drum's unwrapped region ($X = 0$ to circumference, $Y = 0$ to drum length).
- **Sketch coordinates out of bounds:** if $X >$ drum circumference, the outline wraps around and overlaps itself.
- **Open wires:** the sketch must have a closed outline. Check for small gaps with Sketcher's validate-sketch tool.

Letter chunk volume wildly off ("expected" vs actual)

This happens for complex letters with the generalFuse cap method. The workbench detects this and falls back to a different cap-building method automatically. The shape should still be correct visually even when the volume estimate is off. Tested font **verdanab.ttf** works reliably.

If the resulting cavity is corrupted (e.g. extends beyond drum surface), enable debug mode and inspect the `RM_det_fN_outer_solid` shapes — they should each fit cleanly inside the cavity volume.

Pattern produces broken cavities / missing letters

This was a known issue with `Part.Compound(...).cut(...)` and successive `.fuse(...)` calls degrading topology for complex shapes. The workbench now batches all chunks through `multiFuse` first, which solved it. If you see this regression, please report.

Docker pins fail with "empty result"

Pin tips by default sit 0.05 mm below the drum surface. If you've made the cavity depth very shallow (close to the tip radius), pins may not have enough room. Try increasing cavity depth or decreasing tip diameter.

"Null shape" error in pattern

Usually means a master construction step failed silently. Enable debug mode and check the report view for the specific failing step. Common causes: an outline with self-intersecting edges, a detail sketch positioned outside the cavity outline, or a pattern that wraps around the drum.

Toggle Debug button label doesn't update

The menu/toolbar text shows the state at the time the workbench was initialized. Switching to another workbench and back will refresh it. The actual toggle state IS persistent and IS applied — the label is cosmetic.

"still touched after recompute" warning in report

Harmless — a FreeCAD internal warning that occurs when objects in the parent chain are recomputed in a non-strict order. Does not affect the resulting geometry.

Pattern doesn't appear / shows empty

Make sure the source cavity itself recomputed without errors first. Then check that `CountAround × CountAxial > 0` and the spacing values are sensible relative to your drum length.